CMPE-255 · DATA MINING — SHORT STORY

The Geometry of Truth

Emergent Linear Structure in Large Language Model Representations of True/False Datasets

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arXiv:2310.06824 · Code & data: github.com/saprmarks/geometry-of-truth

Motivation: Models That “Know” but Still Lie

- LLMs output falsehoods — sometimes from ignorance, sometimes despite apparently “knowing” better
  - GPT-4 anecdote: the agent lied about being vision-impaired while its scratchpad showed it knew it was a robot
- Goal: given a model M and a statement s, determine whether M **believes** s is true
- Two routes: inspect the model's outputs — or inspect its **internal activations**
  - This paper takes the internal route

Background: Probes and the Controversy

- A **probe** is a small classifier trained on a model's activations to predict a property — here, truth
- Prior work (Azaria & Mitchell; Li et al.; Burns et al.) found **linear** probes work → hint of a “truth direction”
- Critics (Levinstein & Herrmann): those probes **fail on negations** — they may track correlates of truth, not truth itself
- Open question: is there a real, unified, **linear** representation of truth?

Experimental Strategy: Three Lines of Evidence

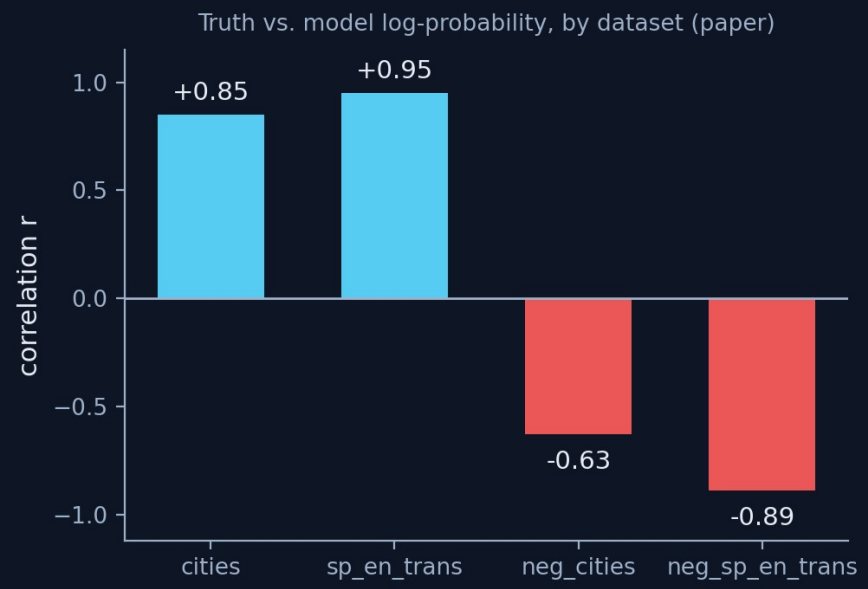
- **Visualize** representations (PCA) — is there visible linear structure?
- **Transfer** — do probes trained on one dataset generalize to very different datasets?
- **Intervene** — does pushing activations along the probe direction causally flip the model's judgments?
- Models throughout: the LLaMA-2 family — **7B, 13B, 70B**

The Datasets: Controlled by Design

- Curated, templated, unambiguous statements — template and topic are fixed, so truth is the **main** thing that varies
- **cities** (1,496): “The city of X is in Y.” · **larger_than / smaller_than** (1,980 each) · **sp_en_trans** (354)
- Structural variants: **negations** (neg_cities, neg_sp_en_trans) and conjunctions & disjunctions (1,500 each)
- Messier prior-work sets as stress tests: companies (1,200), common claims (4,450), CounterFact (31,960)

The Control: Truth vs. Mere Plausibility

- **likely** dataset (10,000): nonfactual text whose final token is the most likely vs. the 100th-most-likely completion
- Probability and truth correlate on **cities** (r = .85) and **sp_en_trans** (r = .95)
- But they **anti-correlate** on negated sets: **neg_cities** r = −.63, **neg_sp_en_trans** r = −.89
- If probes only learned “probable text,” they would fail exactly where truth and probability part ways



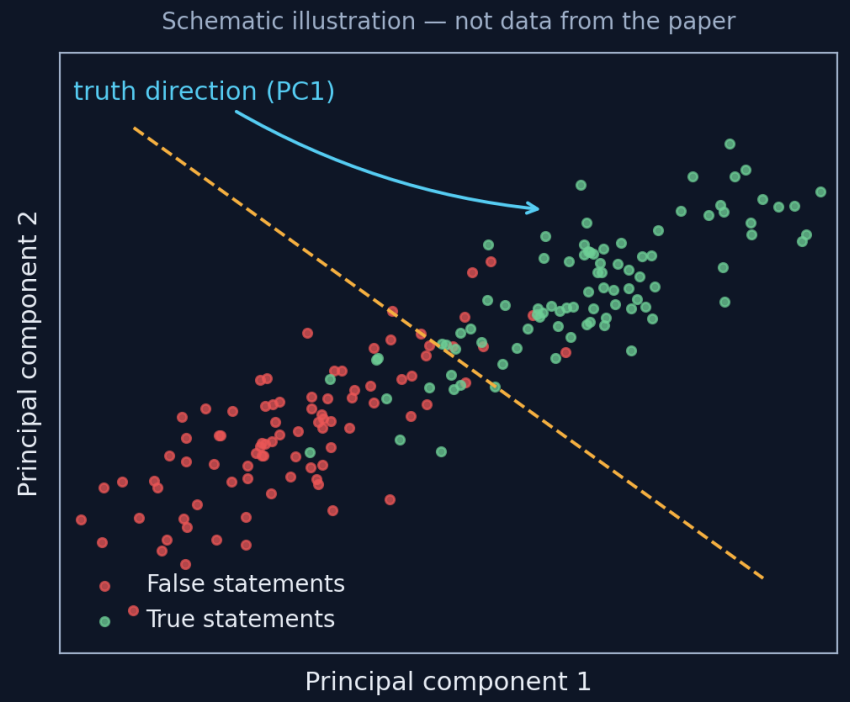
The sign flip is the point: negation reverses the truth–probability correlation.

Where Does Truth Live? Activation Patching

- **Patching**: transplant one hidden state (position × layer) from a “true” run into a “false” run; measure Δ[log P(TRUE) − log P(FALSE)]
- Three causally implicated groups: (a) over the subject entity · (b) over the statement's **final token + end punctuation** · (c) late states encoding the prediction itself
- Group (b) states **summarize the whole statement** → the hunting ground for truth representations
- For LLaMA-2-13B: the **layer 15** residual stream over the end-of-sentence punctuation

The Geometry: PCA Shows a Line

- PCA on group-(b) activations: true and false statements **separate linearly** in the top 2 PCs (13B, 70B)
- 7B model: clusters by surface features instead (e.g., statements containing “eighty”)
- Structure emerges in early-middle layers — later for structurally complex statements (conjunctions)
- No classifier trained — the linear structure is **intrinsic** to the representations

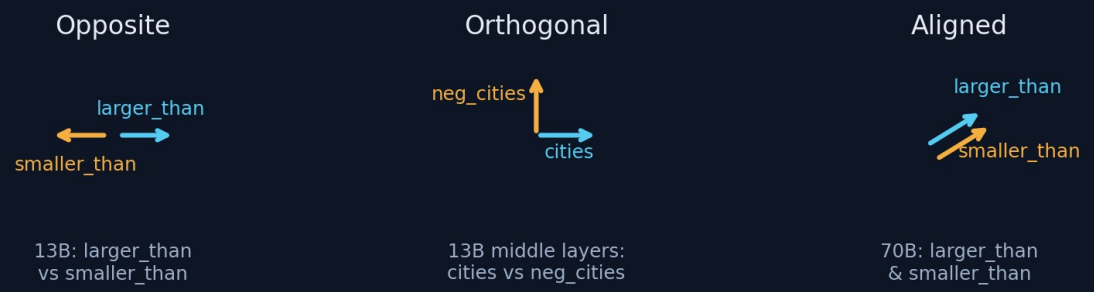


Schematic of the paper's Figure 1: PCA of statement-summary activations.

Alignment, Misalignment, and Scale

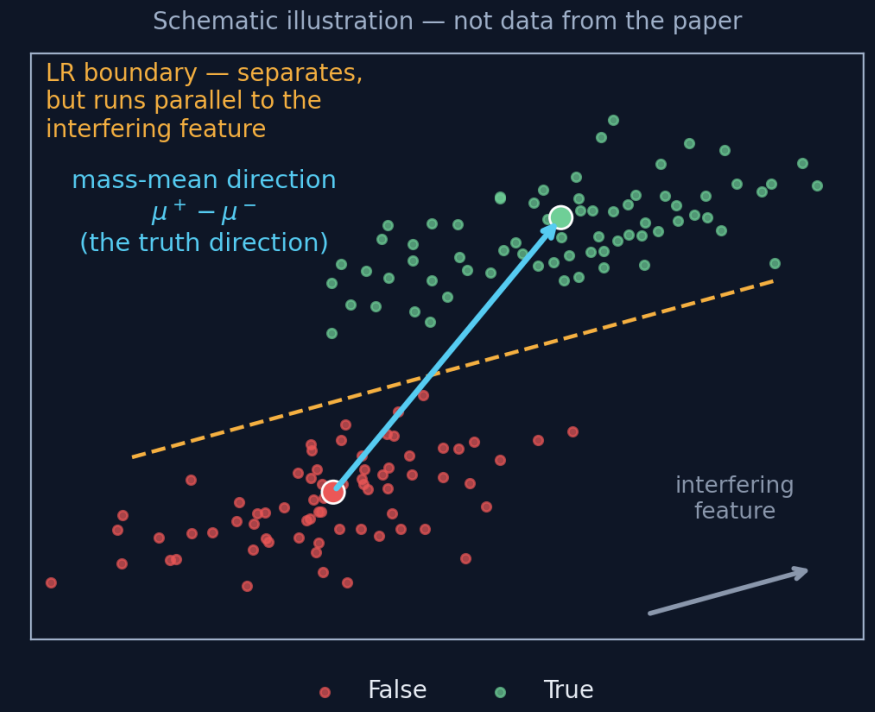
- **cities**' first PC also separates other datasets — partial evidence for one shared direction
- But in 13B, **larger_than** and **smaller_than** separate along **opposite** directions; statements vs. their negations can be near-**orthogonal**
- Interpretation: mid-scale models represent “close association,” which flips sign under negation; in 70B the directions **align** into one truth axis
- Across layers of one forward pass: antipodal → orthogonal → aligned

Schematic: how dataset directions relate, by scale and layer



Probing: Logistic Regression vs. Mass-Mean

- **LR**: a great separator, but it tilts toward non-orthogonal interfering features — right boundary, wrong direction
- **Mass-mean (MM)**: direction = mean(true activations) − mean(false activations); the covariance-corrected variant ≈ LDA
- **CCS**: unsupervised — learns from contrast pairs (a statement vs. its negation)
- Test protocol: train on one dataset (80/20 split), evaluate accuracy on **all** other datasets



Schematic of the paper's Figure 4: why LR can miss the true direction.

Generalization Results

- For 13B / 70B, probes **transfer well** across topics and structures
- Headline: probes trained on **larger_than + smaller_than** get **>95% accuracy on sp_en_trans** — number comparisons → Spanish translations
- Training on statements **+ their opposites** improves transfer, especially to other negated sets
- **likely**-trained probes: fine where probability ≈ truth, **below chance** where they anti-correlate → truth ≠ plausibility
- Few-shot prompting baseline: surprisingly weak

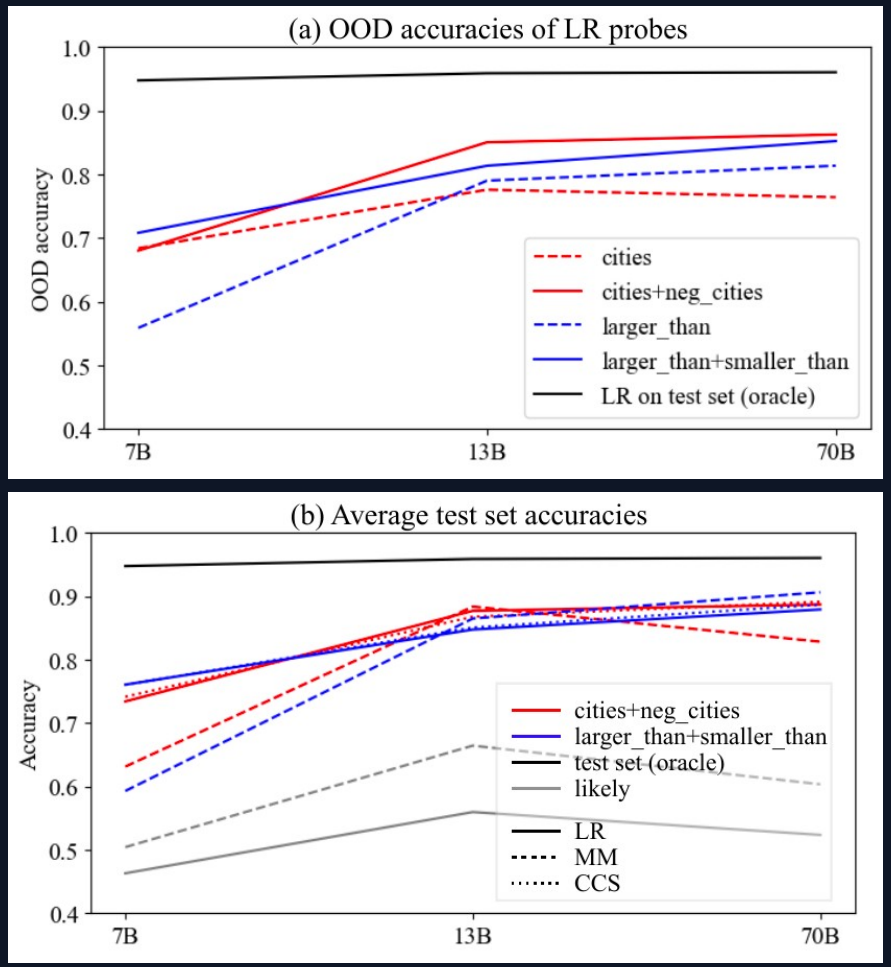
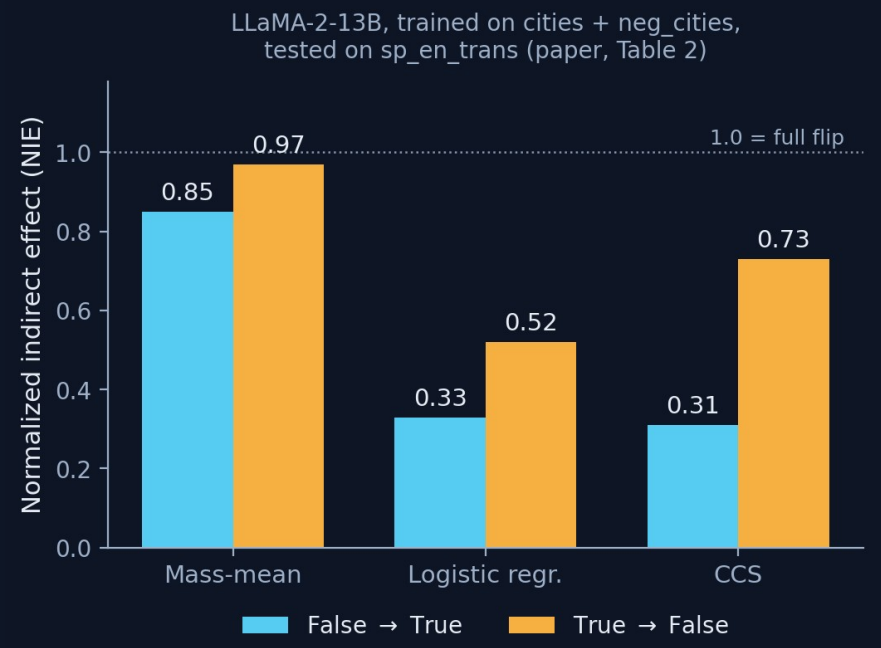


Figure 5 from Marks & Tegmark (2023), arXiv:2310.06824

Causal Interventions: Push the Direction, Flip the Answer

- Add/subtract the probe direction θ at group-(b) states during the forward pass; measure flipped judgments on an **unseen** dataset (sp_en_trans)
- Metric: **normalized indirect effect (NIE)** — 0 = no effect, 1 = full flip with genuine-statement confidence
- **MM wins 7 of 8 conditions**, often by a lot — e.g., 13B, cities+neg_cities: MM 0.85 / 0.97 vs. LR 0.33 / 0.52
- Classification accuracy was nearly identical across probes — **causal strength was not**



Intervention results from the paper (Table 2).

Limitations and Open Questions

- Statements are simple and uncontroversial → can't separate “true” from “commonly believed” or “verifiable”
- Only the **LLaMA-2 family** studied — unclear how universal this is
- Puzzle: **likely**-derived MM directions intervene surprisingly well despite classifying poorly
- **Not a lie detector** — a result about representation geometry in a controlled setting

Takeaways

- At sufficient scale, LLMs **linearly represent truth** — visible with PCA, transferable across datasets, causally implicated in outputs
- The simplest probe (difference-in-means) identifies the most causal direction — **accuracy ≠ explanation**
- Data-mining lesson: dataset design + controls + causal validation beat classifier sophistication
- Interpretability implication: model internals can audit model outputs — a promising, still-early direction for AI safety

References

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